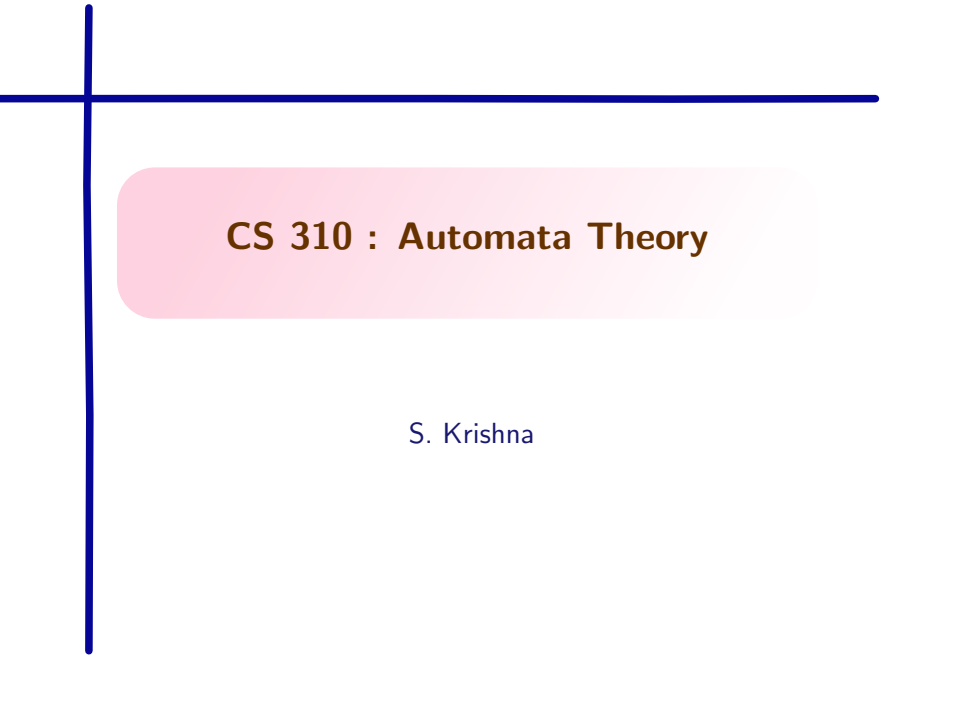


A decorative blue crosshair consisting of a vertical line and a horizontal line intersecting at the top-left corner of the slide.

CS 310 : Automata Theory

S. Krishna

Recall : Context-free grammar (CFG)

Definition

A Context-free grammar(CFG) G is a four-tuple

$$(N, T, P, S)$$

where

- ▶ N is the set of nonterminals
- ▶ T is the set of terminals,
- ▶ P is the set of production rules, and
- ▶ $S \in N$ is the start symbol.

Recall : Language of a grammar

Definition

Let $G = (N, T, P, S)$ be a CFG. The language of G , denoted $L(G)$, is the set of words (with only terminals!) that have derivations from the start symbol.

$$L(G) \triangleq \{w \in T^* \mid S \xRightarrow{G^*} w\}$$

If a language L is the language of some context-free grammar, then we say L is a context-free language (CFL).

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If a language L is the language of some context-free grammar, then we say L is a **context-free language (CFL)**. But we had said, that L is a CFL if it is accepted by a NPDA!

Theorem

$L = L(G)$ for some context-free grammar G iff L is accepted by some NPDA.

Equivalence of CFGs and NPDAs

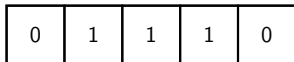
Simulating a CFG with a NPDA

Consider $S \rightarrow 0S0 \mid 1S1 \mid \epsilon \mid 1$

- ▶ Build a NPDA accepting words over $\{0, 1\}$ generated by this grammar
- ▶ Where to store nonterminals in the NPDA?
- ▶ How to simulate the production rules?

Simulating a CFG with a NPDA

$$S \rightarrow 0S0 \mid 1S1 \mid \epsilon \mid 1$$



stack



initial: stack = S, nothing read yet

Simulating a CFG with a NPDA

$S \rightarrow 0S0 \mid 1S1 \mid \epsilon \mid 1$

0	1	1	1	0
---	---	---	---	---



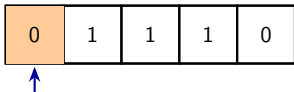
stack

0
S
0

ϵ -transition: guess $S \rightarrow 0S0$

Simulating a CFG with a NPDA

$$S \rightarrow 0S0 \mid 1S1 \mid \epsilon \mid 1$$



stack

0
S
0

match: we will pop the stack top and advance the pointer

Simulating a CFG with a NPDA

$$S \rightarrow 0S0 \mid 1S1 \mid \epsilon \mid 1$$

0	1	1	1	0
---	---	---	---	---



stack

S
0

popped the matched symbol, pointer advanced

Simulating a CFG with a NPDA

$S \rightarrow 0S0 \mid 1S1 \mid \epsilon \mid 1$

0	1	1	1	0
---	---	---	---	---



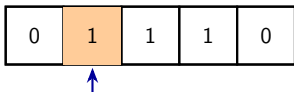
ϵ -transition: guess $S \rightarrow 1S1$

stack

1
S
1
0

Simulating a CFG with a NPDA

$$S \rightarrow 0S0 \mid 1S1 \mid \epsilon \mid 1$$



stack

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stack

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popped the matched symbol, pointer advanced

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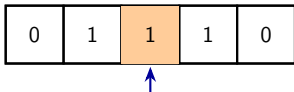
stack

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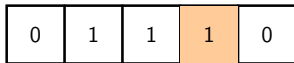
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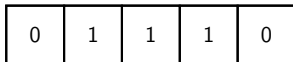
stack

1
0

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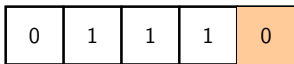
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Simulating a CFG with a NPDA

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0	1	1	1	0
---	---	---	---	---

stack



popped the matched symbol, pointer advanced

Input done, stack empty. Accepted!

The NPDA behind the simulation

Recall the grammar

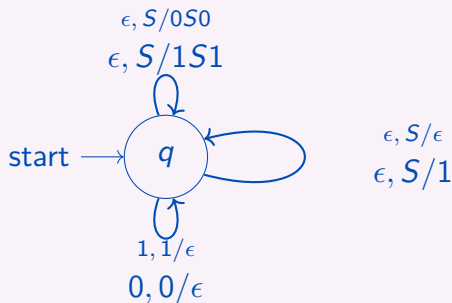
1. $S \rightarrow 0S0$

2. $S \rightarrow 1S1$

3. $S \rightarrow \epsilon$

4. $S \rightarrow 1$

$$A = (\{q\}, \{0, 1\}, \{0, 1, S\}, \delta, q, S, \{\})$$



CFGs to NPDAs

Theorem

Let $G = (N, T, P, S)$ be a CFG grammar. The PDA

$$A = (\{q\}, T, N \cup T, \delta, q, S, \{\})$$

recognizes $L(G)$ by **empty stack**, where δ is defined as follows.

$$\delta(q, \epsilon, B) \triangleq \{(q, \beta) \mid B \rightarrow \beta \in P\}$$

$$\delta(q, a, a) \triangleq \{(q, \epsilon)\}.$$

CFGs to NPDAs

Theorem

Let $G = (N, T, P, S)$ be a CFG grammar. The PDA

$$A = (\underbrace{\{q\}}_{\text{Single state}}, T, N \cup T, \delta, q, S, \{\})$$

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Push anticipated productions

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Verify the anticipated guess while popping

Push anticipated productions

CFGs to NPDAs : Correctness

Given $G = (N, T, P, S)$ be a CFG, construct PDA $A_G = (Q, \Sigma, \Gamma, \delta, q_0, \perp, F)$, where

► $Q = \{q\}$, $q_0 = q$, $\Sigma = T$, $\Gamma = N \cup T$, $\perp = S$, $F = \emptyset$.

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Lemma

$L(G) = L(A_G)$, where acceptance is by empty stack.

► Suppose $w \in L(G)$. Show that w is accepted by A_G .

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- ▶ Consider the **leftmost derivation** for w in G .
- ▶ If the length of the derivation is 1, then $w = a \in T$.

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$L(G) = L(A_G)$, where acceptance is by empty stack.

- ▶ Suppose $w \in L(G)$. Show that w is accepted by A_G .
- ▶ Consider the **leftmost derivation** for w in G .
- ▶ If the length of the derivation is 1, then $w = a \in T$. In this case, $S \rightarrow a$ is a rule in P . Then the initial stack content S will be popped and replaced with a on the stack, and this a is matched with the input a , leading to acceptance. (**Induct on derivation length?**)

CFGs to NPDAs : Correctness

Recall: we want to show $w \in L(G) \Rightarrow w$ accepted by A_G .

- To make induction possible, we prove something more general: for every $\alpha \in \Gamma^*$ and $w \in T^*$,

$$\alpha \Rightarrow_{lm}^n w \implies (q, w, \alpha) \vdash^* (q, \epsilon, \epsilon).$$

Taking $\alpha = S$ recovers the claim we want.

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Taking $\alpha = S$ recovers the claim we want.

- ▶ We induct on n , the number of steps in the leftmost derivation.

CFGs to NPDAs : Correctness (base cases)

- **Base case** ($n = 0$): $\alpha \in T^*$ already, so $\alpha = w$, and A_G matches it against itself symbol-by-symbol via $\delta(q, a, a) = (q, \epsilon)$.

CFGs to NPDAs : Correctness (base cases)

- ▶ **Base case** ($n = 0$): $\alpha \in T^*$ already, so $\alpha = w$, and A_G matches it against itself symbol-by-symbol via $\delta(q, a, a) = (q, \epsilon)$.
- ▶ **Base case** ($n = 1$): $\alpha = wAw'$ with $w, w' \in T^*$, $A \in N$, and $A \rightarrow y \in P$ for some $y \in T^*$. Then the derived string is wyw' , and

$$(q, wyw', wAw') \vdash^{|w|} (q, yw', Aw') \vdash (q, yw', yw') \vdash^{|y|+|w'|} (q, \epsilon, \epsilon),$$

matching w symbol-by-symbol, then using $(q, y) \in \delta(q, \epsilon, A)$ to pop A and push y , then matching yw' symbol-by-symbol.

CFGs to NPDAs : Correctness (continued)

- ▶ **Inductive step** ($n > 1$): write $\alpha = x\beta$ where $x \in T^*$ is the longest terminal prefix of α , so $\beta = A\gamma$ begins with a nonterminal A .

CFGs to NPDAs : Correctness (continued)

- ▶ **Inductive step** ($n > 1$): write $\alpha = x\beta$ where $x \in T^*$ is the longest terminal prefix of α , so $\beta = A\gamma$ begins with a nonterminal A .
- ▶ Terminals are never rewritten, so none of the n steps touch x : we get $w = xw'$ with $\beta \Rightarrow_{lm}^n w'$.

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 $(q, w', \delta\gamma) \vdash^* (q, \epsilon, \epsilon)$.

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$$(q, w, \alpha) = (q, xw', x\beta)$$

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where the first $|x|$ moves match x symbol-by-symbol via $\delta(q, a, a)$, the next move uses $(q, \delta) \in \delta(q, \epsilon, A)$, and the rest is the IH.

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$$(q, w, \alpha) = (q, xw', x\beta) \vdash^{|x|} (q, w', A\gamma) \vdash (q, w', \delta\gamma) \vdash^* (q, \epsilon, \epsilon)$$

where the first $|x|$ moves match x symbol-by-symbol via $\delta(q, a, a)$, the next move uses $(q, \delta) \in \delta(q, \epsilon, A)$, and the rest is the IH.

- ▶ This proves the general claim; setting $\alpha = S$ gives $w \in L(G) \Rightarrow w$ accepted by A_G .

CFGs to NPDAs : Converse

- ▶ From A_G to G . That is, whenever the constructed NPDA A_G accepts a word by empty stack, the word is generated in G . How to proceed?
- ▶ Let us relook thru an example again